MathEXplainedMAGAZINE

Mathematical Beauties within Chess

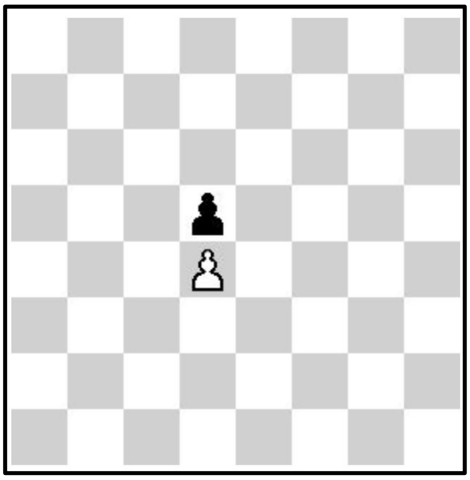
By Harshil N.

You might have played a chess game before. It is great fun really – arranging your pieces, capturing opponents' pieces, creating checkmates, etc… However, did you know that there is significant mathematical depth in chess? Today we will be talking about two different areas in chess where mathematics finds a way to fit in.

What is the number of different games that can occur when playing chess? Let's first analyze the opening move; since both players in chess have 20 possible opening moves (2 moves for each of the 8 pawns as a pawn can move one step or two steps in the first move, and 2 moves for each of the two knights), when both players finish their first move, a total of 20*20 = 400 possible board positions can be made. But as you know, the opening move is only a small fraction of the game. Extending the last example, how do we calculate the number of different games of chess that can be played? This was calculated by Claude Shannon and the number is referred to as the Shannon Number – 10^120. How do we even get this number? For this result, approximations are helpful: there are about 10^3 possibilities for a complete move (i.e. both players move) and a chess game has approximately 40 total moves; hence, from the common multiplication principle of combinatorics, the Shannon number is given. Think about that number for a second, how big is it? The age of our universe is on a magnitude scale of 10^18 and the number of atoms in our universe is 10^80. Thanks to the Shannon number, we can quantify how complex of a game Chess truly is.

We can also study chess in a combinatorial game theory (CGT) lens. Let's start with simple CGT terminology: a 0 game (quite like 0 numerically) is a game in which neither player can move. There are different outcome classes of a game, but a game G has an outcome class as P if the first player who plays loses the game. Finally, a fundamental result from CGT states that all 0 games have an outcome class as P (in particular, the statement is not bi-directional).

Historically, studying chess in CGT is a difficult task due to strict rules that CGT studied games need to offer. However, slightly modifying or loosening the rules of chess can make room for Chess Theory. Here, we can analyze when the board is limited to pawns. As we do with many different combinatorial games, we need to understand how to build up numbers. Let's start with the 0 as shown in the image.



Notice the board configuration if a player starts, they will lose as they have no option (since pawns can only capture diagonally). Hence, the outcome class for this game is first-player to move loses, or in CGT a P game, and hence a 0 game.

We can continue in a similar fashion to talk about different numbers, like 1, -1, and even some infinitesimal values! However, this requires some more CGT understanding. However, we hope we excited you into how chess can actually be studied in this field.

Today, we talked about a very simplistic push into some mathematics behind chess which is pretty uncommon. We hope now you have a little more appreciation for chess apart from its face value.

Pi Day Math Crossword

By Kevin C.

1.	2.
3.	

ONE DOWN

$\sum_{n=1}^{\infty} \frac{1}{n^2(n+1)^2}$ can be written as $\frac{\pi^a}{b} - c$ where a, b, and c are all integers. Evaluate $a(b+c)$.

THREE ACROSS

The number x can be written as $\frac{a}{b}$ where a and b are relatively prime positive integers. Given that $\frac{\pi}{2} = 2\arctan(\frac{1}{4}) + \arctan(x)$, evaluate $a+b$.

ONE ACROSS

Consider an infinite sequence of circles as follows:

1. The first circle has a radius of 1 and is centered at the origin.
2. Each subsequent circle is scaled by a factor of $\sqrt{2-\sqrt{2}}$ relative to the previous circle and positioned so that its center passes through the circumference of the previous circle.
3. The centers of all circles are collinear.

Given that the area enclosed by the sequence of circles can be written as x, compute the integer closest to $2x$.

TWO DOWN

Consider the sequence a_n with $a_0 = \pi$ and $a_n = \lfloor \pi * a_{n-1} \rfloor$ for all $n \geq 1$. Determine $\lfloor \frac{a_3}{2} \rfloor$.